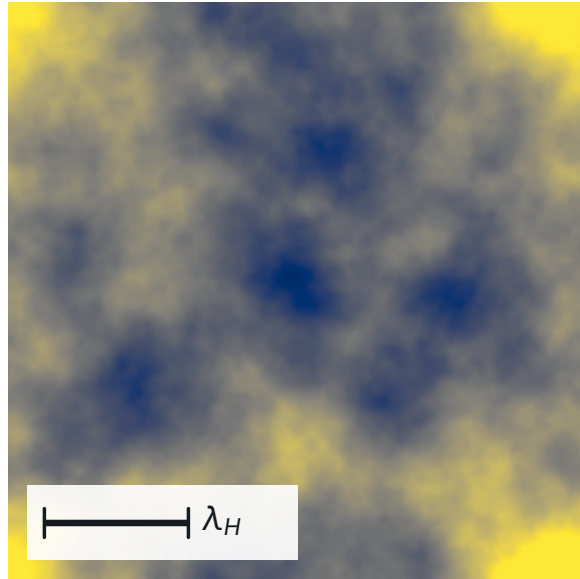


Host mean trait

disperses far: varies over a long distance



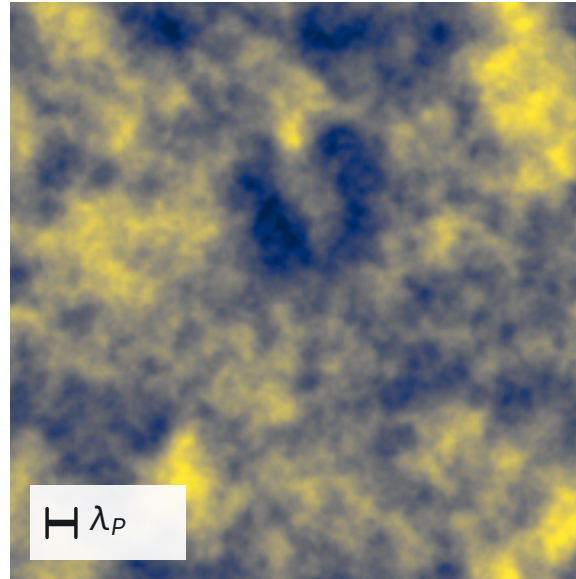
spatial scale

$$\lambda = \sigma / \sqrt{GA}$$

variation among places $V = 1 / (NA \sigma^2)$

Parasite mean trait

disperses little: varies over a short distance

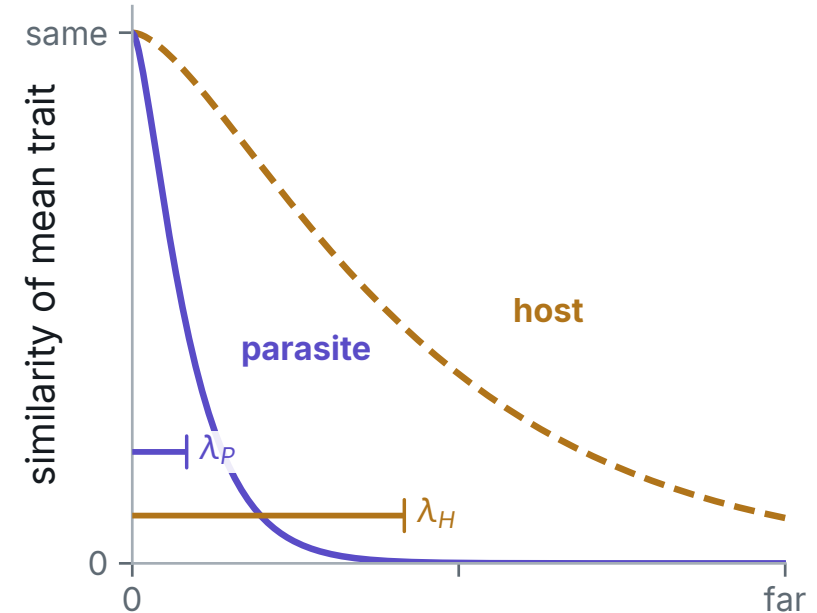


dispersal distance, over the root of genetic variance times abiotic selection

more in small populations (drift), less with strong selection and wide dispersal

Each species has its own spatial scale

λ : the distance over which mean traits stay alike



distance between two places

one scale for each species; and between them, a cross-covariance at every distance, made by biotic selection